

Autonomous Computer Vision Based Human-Following Robot

1. Problem Statement

Autonomous human-following robots require a vision system that can detect a person, estimate the person's position relative to the camera, and convert this information into movement decisions. The objective of this project is to develop a computer-vision-based human-following robot prototype that detects a human using an onboard-camera image, estimates approximate position and distance, analyzes the visual environment, and generates a corresponding simulated robot command. The project is implemented as a software prototype without physical robot hardware.

2. Dataset

The original project dataset contained 95 image files, including 60 JPG files and 35 PNG files. Many were processed or segmented versions of the same images. To avoid treating derived images as independent samples, the experiments used the 18 clean original JPG images as the primary dataset. These images contain people in different positions and apparent distances from the camera. A separate video dataset containing crosswalk.avi, fourway.avi, and night.avi with corresponding CSV files was also examined for additional testing.

3. Proposed Methodology

The proposed pipeline is: Input Image → YOLO11n Human Detection → Bounding Box Analysis → Position & Distance Estimation → Environment Analysis → Decision Logic → Simulated Robot Command. A pretrained YOLO11n detector identifies the person. Bounding-box center determines left, center, or right position, while bounding-box height provides an approximate NEAR, MEDIUM, or FAR distance category. K-Means clustering was retained as an auxiliary image-segmentation component. An environment-aware safety module uses mean grayscale brightness and Canny edge ratio to classify scenes as NORMAL, LOW LIGHT, or DARK / UNCLEAR.

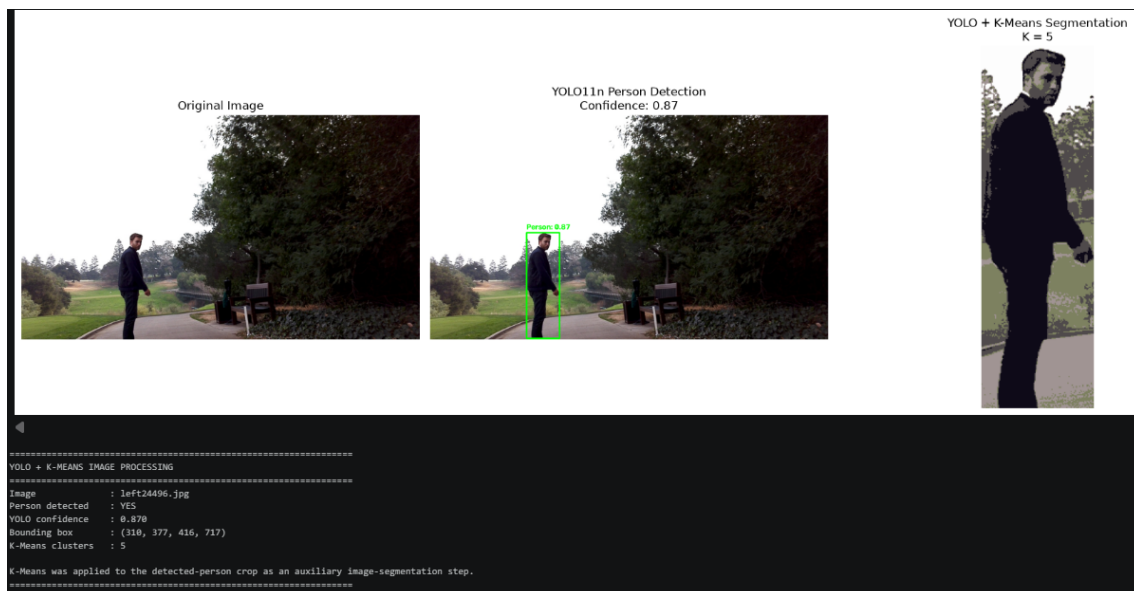


Figure 1: Original image, YOLO11n person detection (confidence 0.87), and K-Means segmentation (K = 5) of the detected-person crop for left24496.jpg.

4. Implementation

The system was implemented in Python using OpenCV, NumPy, Matplotlib, and Ultralytics YOLO. A rule-based decision layer converts visual measurements into simulated commands. The decision hierarchy gives priority to safety: no person → STOP - NO PERSON; NEAR → STOP; LEFT → TURN LEFT; RIGHT → TURN RIGHT; FAR and centered → MOVE FORWARD; otherwise → FOLLOW. The environment module can modify the command when lighting conditions indicate that the scene is unclear.

ENVIRONMENT-AWARE SAFETY MODULE			
POSITION	DISTANCE	ENVIRONMENT	FINAL COMMAND
CENTER	MEDIUM	NORMAL	FOLLOW
LEFT	MEDIUM	NORMAL	TURN LEFT
RIGHT	MEDIUM	NORMAL	TURN RIGHT
CENTER	FAR	NORMAL	MOVE FORWARD
CENTER	FAR	LOW LIGHT	MOVE FORWARD - CAUTIOUS
CENTER	MEDIUM	DARK / UNCLEAR	STOP - ENVIRONMENT UNCLEAR
CENTER	NEAR	NORMAL	STOP

Figure 2: Environment-aware safety module — final command for each combination of position, distance, and environment.

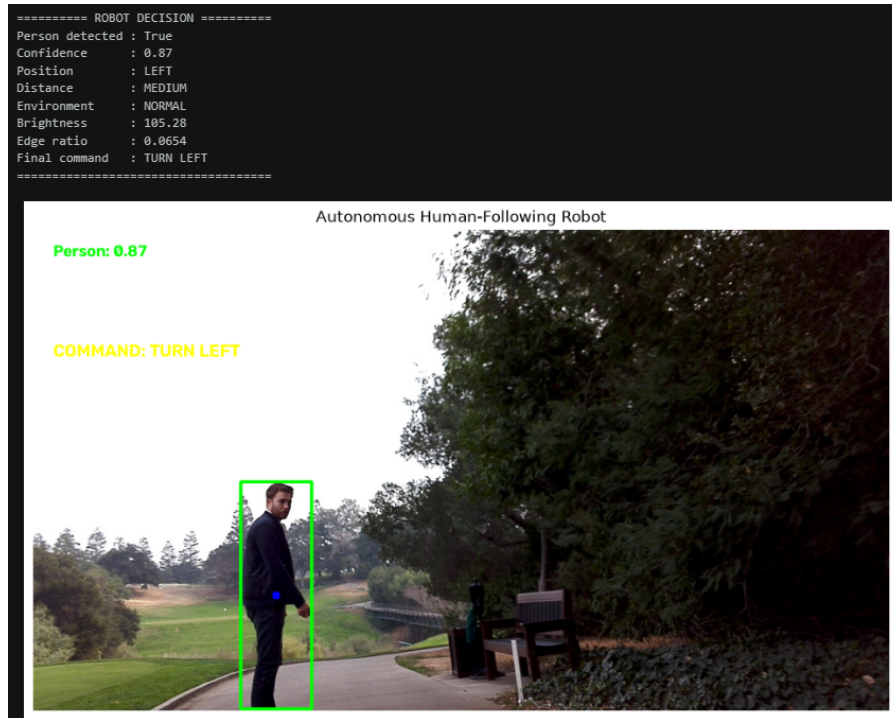


Figure 3: Sample robot decision output — person detected (0.87), LEFT, MEDIUM distance, NORMAL environment, command TURN LEFT.

5. Experimental Results

Metric	Result
Images tested	18
Persons detected	18
Images with no detection	0
Detection rate	100%
Average detection confidence	0.890

All 18 images produced a valid person detection. The 100% detection rate should not be interpreted as 100% accuracy because the dataset is small and does not contain an independently annotated evaluation set.

Command	Count
FOLLOW	5
STOP	8
TURN LEFT	1
TURN RIGHT	3
MOVE FORWARD	0
STOP - ENVIRONMENT UNCLEAR	1

One image (left24875.jpg) was classified as DARK / UNCLEAR and generated STOP - ENVIRONMENT UNCLEAR. The command counts represent system behavior, not command accuracy, because no ground-truth robot commands

were available.

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FINAL HUMAN-FOLLOWING ROBOT EVALUATION
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IMAGE	CONF	POSITION	DISTANCE	ENVIRONMENT	COMMAND
left0445.jpg	0.847	CENTER	MEDIUM	NORMAL	FOLLOW
left0586.jpg	0.917	CENTER	NEAR	NORMAL	STOP
left21281.jpg	0.934	CENTER	NEAR	NORMAL	STOP
left22088.jpg	0.950	LEFT	NEAR	NORMAL	STOP
left22156.jpg	0.845	CENTER	MEDIUM	NORMAL	FOLLOW
left22331.jpg	0.936	LEFT	NEAR	NORMAL	STOP
left22741.jpg	0.797	RIGHT	FAR	NORMAL	TURN RIGHT
left23075.jpg	0.885	RIGHT	MEDIUM	NORMAL	TURN RIGHT
left23590.jpg	0.869	CENTER	MEDIUM	NORMAL	FOLLOW
left23761.jpg	0.851	RIGHT	MEDIUM	NORMAL	TURN RIGHT
left23945.jpg	0.889	CENTER	NEAR	NORMAL	STOP
left24496.jpg	0.870	LEFT	MEDIUM	NORMAL	TURN LEFT
left24720.jpg	0.907	CENTER	MEDIUM	NORMAL	FOLLOW
left24875.jpg	0.932	CENTER	NEAR	DARK / UNCLEAR	STOP - ENVIRONMENT UNCLEAR
left25057.jpg	0.916	CENTER	NEAR	NORMAL	STOP
left25196.jpg	0.922	CENTER	NEAR	NORMAL	STOP
left25343.jpg	0.922	CENTER	NEAR	NORMAL	STOP
left9236.jpg	0.874	CENTER	MEDIUM	NORMAL	FOLLOW

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...
STOP : 8
TURN RIGHT : 3
TURN LEFT : 1
STOP - ENVIRONMENT UNCLEAR : 1
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings..
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Figure 4: Final evaluation over all 18 images — confidence, position, distance, environment, and command per image, with command counts.

6. Fine-Tuning Experiment

A fine-tuning experiment used 12 training images, 3 validation images, and 3 test images. Labels were generated using the pretrained YOLO11n detector and visually checked, so they are pseudo-labels rather than independently annotated ground truth.

Model	Precision	Recall	mAP50	mAP50-95
Pretrained YOLO11n	0.986	1.000	0.995	0.995
Fine-tuned YOLO11n	0.003	1.000	0.913	0.523

The pretrained model produced stronger results on this small test set, so it was retained for the final system. The limited pseudo-labeled dataset was insufficient to establish improvement through fine-tuning.

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YOLO11n FINE-TUNING EXPERIMENT
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Dataset split:
Training images : 12
Validation images : 3
Test images : 3

PRETRAINED YOLO11n
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Precision : 0.986
Recall : 1.000
mAP50 : 0.995
mAP50-95 : 0.995

FINE-TUNED YOLO11n
-----
Precision : 0.003
Recall : 1.000
mAP50 : 0.913
mAP50-95 : 0.523

FINAL MODEL SELECTION
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Selected model : Pretrained YOLO11n
Reason : Stronger test-set performance

Note: Fine-tuning used a small pseudo-labeled dataset.
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Figure 5: Fine-tuning experiment output — dataset split, pretrained vs. fine-tuned metrics, and final model selection.

7. Video Testing

Video	Sampled Frames	Detections	Detection Rate
Crosswalk	76	76	100.00%
Fourway	257	252	98.05%
Night	113	107	94.69%

These are detection rates on sampled frames, not full-frame accuracy. Detection remained high across the three scenes, with the lowest sampled-frame rate occurring in the night sequence.

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VIDEO HUMAN-DETECTION EXPERIMENT
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VIDEO      AI FRAMES  DETECTIONS  DETECTION RATE
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Crosswalk   76          76         100.00%
Fourway     257         252        98.05%
Night       113         107        94.69%
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Note: Detection rate is calculated on sampled AI frames.
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Figure 6: Video human-detection experiment output (detection rate computed on sampled frames).

8. Limitations

The final image dataset contains only 18 clean original images, which is too small to establish broad generalization. Fine-tuning used pseudo-labels rather than independent ground truth. Distance estimation is based on bounding-box height and is therefore approximate. The environment module uses brightness and edge information and is not a complete terrain-classification system. K-Means is auxiliary and its contribution to detection performance was not quantitatively established. Robot commands are simulated because no physical hardware was used.

9. Conclusion

This project developed a software prototype for an autonomous computer-vision-based human-following robot. The final pipeline combines pretrained YOLO11n human detection, bounding-box-based position and distance estimation, K-Means image segmentation, environment-aware safety analysis, and rule-based simulated robot control. On the 18 clean original images, YOLO11n detected a person in all 18 cases, with average confidence of approximately 0.890. Additional video experiments demonstrated high person-detection rates. Fine-tuning did not outperform the pretrained model, so the pretrained model was retained. Future work would require a larger independently annotated dataset, improved terrain/traversability classification, better distance estimation, and physical robot integration.

References: YOLO11n/Ultralytics implementation; Golf Bag Carrier Robot Computer Vision project reference material; UE24CS352A Machine Learning Mini-Project Assignment Guidelines and Instructions.